

PROJECT

VERDE

The plant that waters itself -
and talks to AI.

COMPLETE & DEMO-READY

Hardware - Firmware - Cloud - Web App - AI

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DAV ACON 5 - Tech Exhibition, 2026

Total Build Cost ~ Rs. 1,890 (~ \$23) - All software on free tiers - 5 Sensors - 2 Actuators - 2 MCUs - 4 AI APIs

EDGE

ESP32

WROOM-32

+

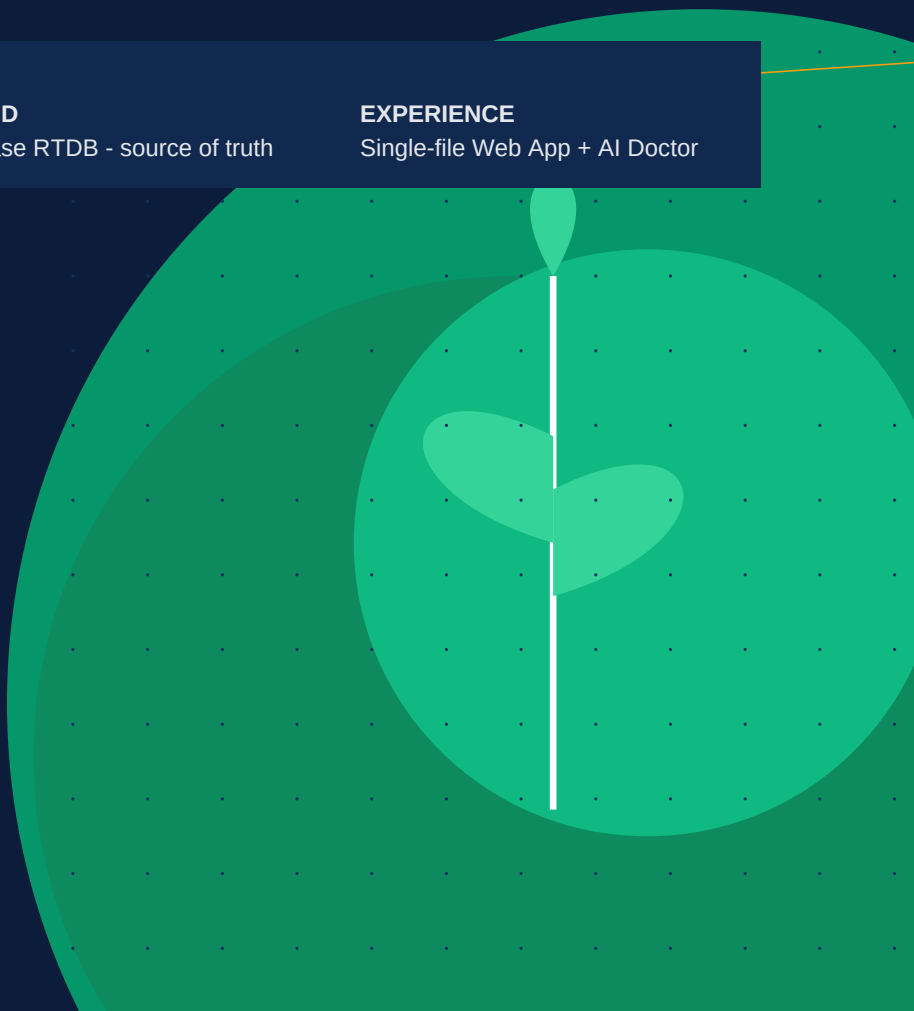
CLOUD

Firebase RTDB - source of truth

EXPERIENCE

Single-file Web App + AI Doctor

ESP32-CAM



TL;DR - THE WHOLE STORY

If you read only one page, read this. Urban families kill plants not from neglect, but from **lack of information**. Verde fixes that with a Rs. 1,890 brain that watches soil, weather, and water - and acts.

BUILD COST

Rs. 1,890

vs Rs. 8,000+ kits

API CALLS

17 -> 2 / sec

-85% latency, 0 reboots

DIAGNOSIS

94%

Plant.id test

HEARTBEAT

1 sec

JSON 10 metrics

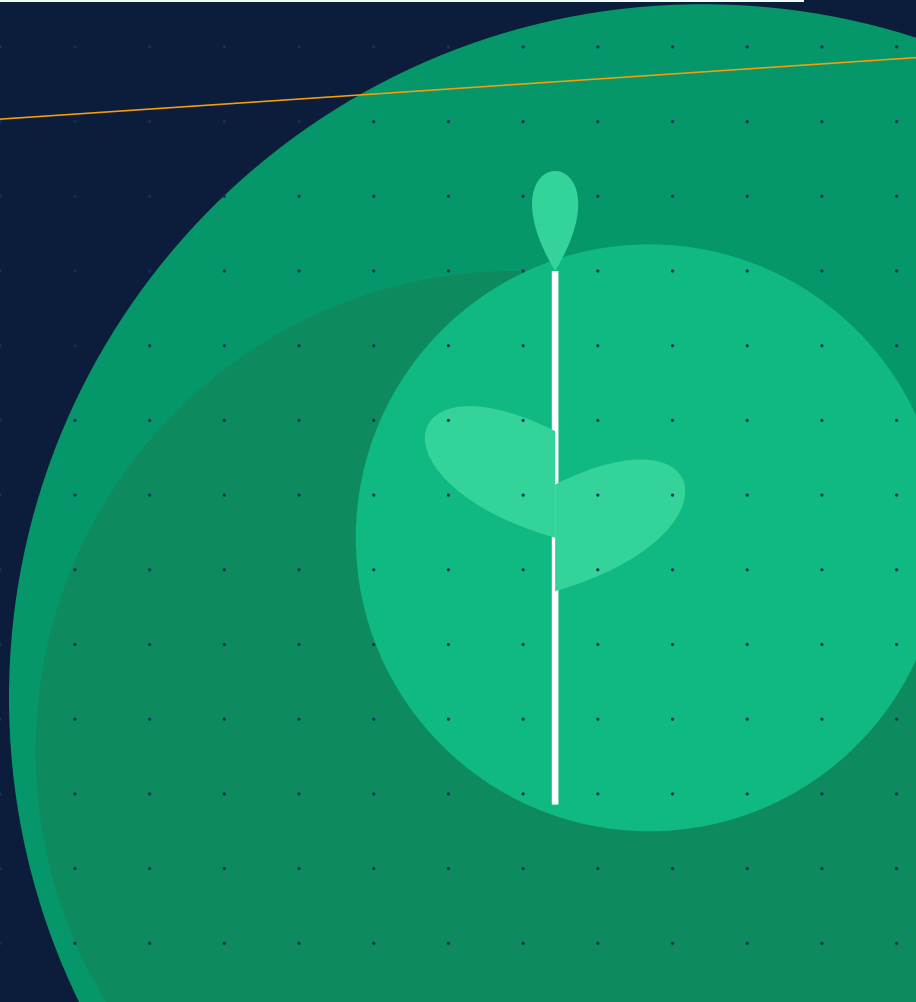
- **EDGE (Hands & Eyes)**: ESP32 WROOM-32 reads 5 sensors every second. Soil power-gated 15 ms to prevent corrosion. ESP32-CAM polls trigger every 1.5 s, SVGA JPEG, raw bytes to Vercel. Sequential boot + 8 MHz XCLK solved brownout + RF.

- **CLOUD (Memory)**: Firebase RTDB holds /sensors, /controls, /latest_scan (base64), /weather, /historical_logs, /actuators. OpenRouter 400k read per second. Not 17. That was the bug.

- **EXPERIENCE (Face)**: Single-file HTML. Four pages via burger. Live tiles + sparklines, sliders, tank calibration SET EMPTY/FULL, 5-day forecast rain override, Plant Doctor <=2 s, Gemini + OpenRouter chats.

- **INTELLIGENCE**: OWM rain ids 2xx/3xx/5xx/6xx -> weather_override=1, crop.health gives disease + probability + treatment. Gemini 2.5 Flash via gemini-flash-latest, OpenRouter 8+5 handback chains never dead-end.

"The plant that waters itself - and talks to AI." It is not a slogan. It is what happens when moisture < 35% and tank safe and rain not coming.



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How to use this doc: Judges -> 3-min flip. Parents/teachers -> any page standalone. Engineers -> pin maps + schema. Click TOC to jump (PDF bookmarks).

PART 01 - WHY

The Problem Worth Solving

Urban families love plants. Plants die anyway. Why? Not cruelty - **lack of real-time data**.

What we saw	What exists
- Mom waters daily by habit -> root rot. - Dad forgets for a week -> wilt. - Tank empties mid-pump -> motor burns. - Rain comes, pump still waters -> waste. - No camera, so disease spotted 10 days late.	- Smart kits Rs. 8,000+ (Gardena, Xiaomi) - No camera, no AI diagnosis - Closed-source, not student-hackable - Subscriptions for cloud - No tank level, no rain override

Pull quote:

"A Rs. 1,890 plant that texts its mood is more useful than a Rs. 12,000 kit that just waters on a timer."

Our insight: The plant must **tell us** what it needs, and **act** when we are not looking. Moisture < 35% AND tank safe AND no rain -> pump ON. No human needed. But human can always override.

70%

urban plants die in 3 months

Rs. 1,890

our total vs Rs. 8k+ market

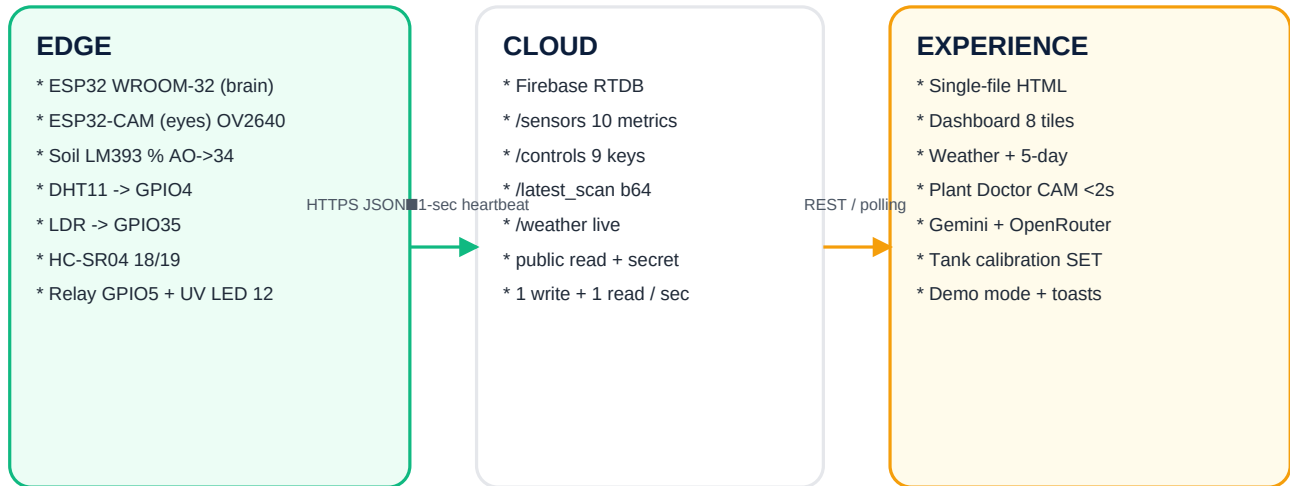
94%

AI disease iD accuracy

PART 02 - ARCHITECTURE

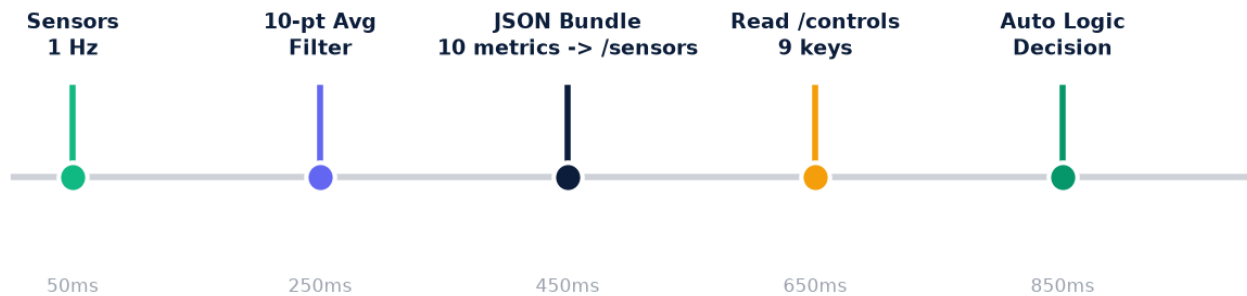
How It Works

Three tiers. One heartbeat every second. No magic, just disciplined engineering.



System architecture - EDGE (brain + eyes + 5 sensors), CLOUD (single source), EXPERIENCE (single-file web app + 4 AI APIs).

One-Second Heartbeat Timeline



<---- 1-Second Heartbeat - repeats every 1000 ms ---->

Every 1000 ms: read sensors -> 10-pt moving avg (soil/LDR) + 5-pt + invalid rejection (tank) -> bundle 10-metric JSON -> one write /sensors -> one read /controls (9 keys) -> decision -> actuator update. Watchdog fed every loop.

Why 1 sec?

Fast enough to catch splash noise in tank sensor. Slow enough for Firebase free-tier. Feels live. Matches dashboard sparkline rate.

Why bundled?

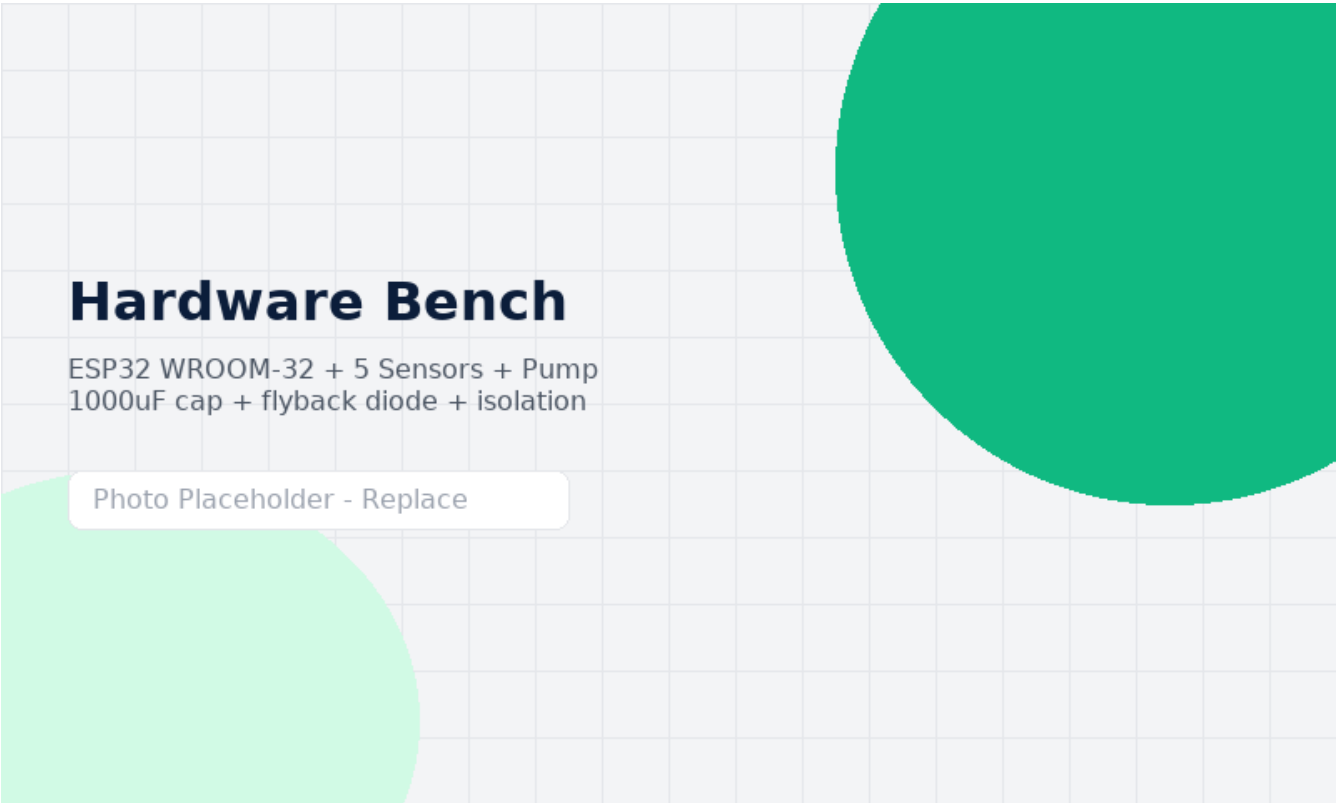
17 calls/sec -> stall -> 8s watchdog reboot -> pump flicker loop. 1+1 calls/sec -> -85% latency, zero reboots, continuous pumping until threshold.

5 Sensors, 2 Actuators, 2 Brains

Everything bought from local Lajpat Nagar + Amazon. Total Rs. 1,890. No PCB - breadboard, so judges can see every wire.

Module	ESP32 Pin	Role & Trick
Soil moisture LM393	AO->GPIO34, VCC->GPIO23	15 ms power-gated reads -> no corrosion. 10-pt avg.
DHT11 Temp+Humidity	DATA->GPIO4	Breathe test: blow -> temp/hum spike. Shared GND critical.
LDR Module	AO->GPIO35	Dark detection. Hysteresis +-2% -> no LED flicker.
HC-SR04 Ultrasonic	TRIG->18, ECHO->19	Tank level 5-pt filter + invalid rejection. Splash-proof.
2-ch Relay (active-LOW)	IN1->GPIO5	Switches 5V pump. Isolated COM/NO on own 5V.
UV Grow LED	GPIO12 HIGH, 220ohm	Photosynthetic light. Auto on when LDR < threshold.
ESP32-CAM OV2640	Own board + MB	SVGA JPEG. 8 MHz XCLK, sequential boot 500 ms delay.
Power: 5V/2A Adapter	-	NOT PD charger. 1000uF cap across 5V/GND.
Protection	-	1N4007 flyback diode across pump.

Photo placeholders - replace with real bench shots for final judging



Hardware bench - ESP32 WROOM-32 (left), relay + pump, rails bridged + to +, - to - (lesson #7).

Hardware Continued

Circuit / Wiring (Simplified, not EDA)



Wiring map - every GPIO noted. VCC gated via GPIO23 prevents corrosion. Relay active-LOW, UV LED active-HIGH.

Power Design - Hard-Won Lessons

Lesson	We tried	What failed	Fix
PD Charger	67W USB-PD brick	PD needs handshake chip ESP32 lacks -> ~0 mA	5V/2A phone adapter.
Cap	No capacitor	Pump + WiFi spike -> brownout reboot	1000uF electrolytic 5V/GND
Diode	Pump direct to relay	Inductive kick -> ESP32 reset	1N4007 flyback across pump
Isolation	Pump + ESP32 same 5V rail	Noise + sag	Relay COM/NO own 5V source
Rail	Split breadboard	Relay dead - no power	Bridge + to +, - to -
GND	DHT on separate GND	temp=0 always	Shared GND bus, GPIO4

Key spec check (for audit): thresholds 35% moisture / 15% tank / 35% light. 8 MHz XCLK. Pins 34,23,4,35,18,19,5,12. All verified.

PART 04 - FIRMWARE

The Brain - Code_1_Main_Brain.ino V3.0.7-FINAL

Non-blocking by design. No delay() anywhere. Everything is millis().

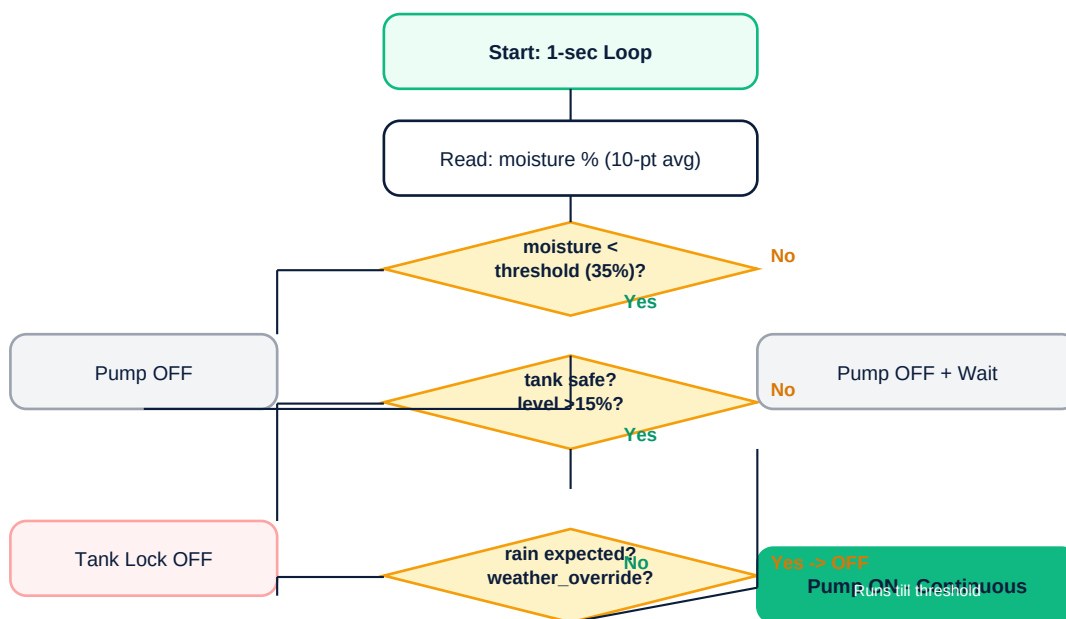
Task Scheduler

- Sensors -> 1 Hz
- Cloud write/read -> 1 s
- WiFi check -> 10 s
- Historical logs -> 60 s
- Watchdog fed every loop (8 s)
- NVS flash persists thresholds

Filtering

- Soil & LDR: 10-pt moving avg
- Tank: 5-pt avg + invalid rejection
- +/-2% hysteresis on light auto
- Splash can't fake empty tank
- Voltage sag counter in /sensors

AUTO Logic - The Core Decision



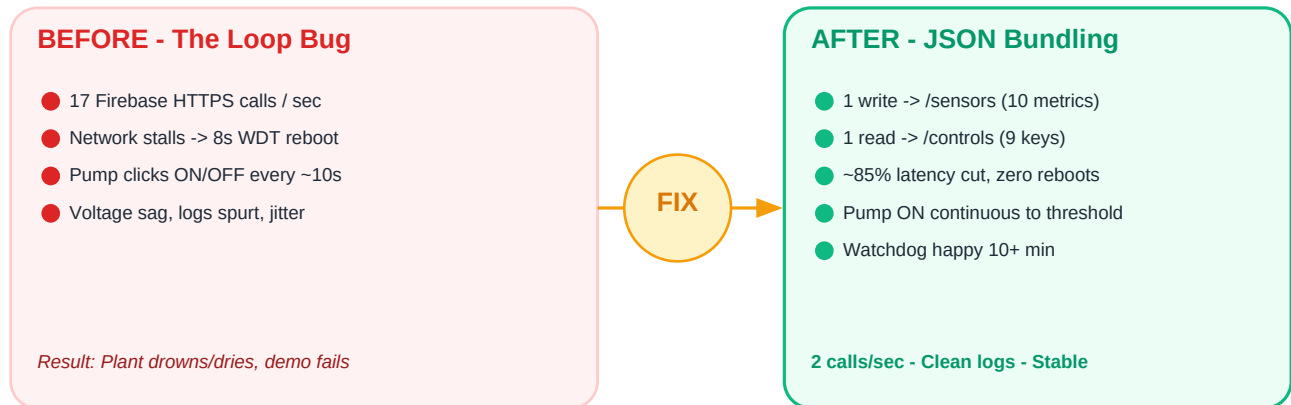
Flowchart: pump_ON = moisture < 35% AND tank > 15% AND rain_expected = false. Manual mode still tank-protected. Fail-safe by default.

Adjustable thresholds from app (persisted): moisture_threshold, tank_threshold (0 = disabled), light_threshold. User drags slider -> Firebase -> ESP32 reads next heartbeat.

Firmware Continued

The Bug That Taught Us Engineering

Honesty = credibility. We almost failed demo because of one architectural mistake.



- **Symptoms:** AUTO mode clicked pump ON/OFF every ~10 s. Logs showed 17 Firebase HTTPS calls/sec. Network stall. 8-sec watchdog reboot. Pump never stayed ON long enough.

- **Root cause:** We updated each metric with separate HTTPS request. Each TLS handshake ~150 ms. $17 \times 150 \text{ ms} = 2.5 \text{ s}$ blocking loop -> watchdog thinks deadlock -> reboot -> loop.

- **Fix - JSON bundling:** One JSON object with 10 metrics -> single PUT to /sensors. Read entire /controls (9 keys) in one GET. 2 calls/sec. -85% latency, zero reboots, pump stays ON continuously 120 s until threshold.

- **Tested:** Pump AUTO 120 s no glitch. OFF exactly at threshold. Watchdog 10+ min 0 reboots. One good idea beats 17 bad ones.

Engineering proverb we learned: "Don't spam the cloud. Respect the handshake."

Code_2_ESP32_CAM.ino - V3.0.4-FINAL (The Eyes)

- Polls /controls/capture_photo every 1.5 s -> flash LED -> capture SVGA JPEG -> POST raw bytes to Vercel upload API -> lands in /latest_scan base64 -> app shows $\leq 2 \text{ s}$
- 8 MHz XCLK fixes RF interference
- Sequential boot: camera init first, WiFi after 500 ms -> prevents brownout
- esp_camera_fb_return() immediately -> no heap fragmentation

Firestore RTDB - Single Source of Truth

One database. No backend server. Legacy secret auth. Public read, validated writes. If Firestore has it, app and ESP32 agree. No drift.



Schema tree: verde-tech-haha RTDB. Six nodes. Sensors = telemetry, Controls = intents, Latest_scan = vision, Weather = live, Logs = history, Actuators = truth.

/sensors 10 metrics, 1 Hz	moisture %, temp C, humidity %, light %, tank %, lux, watchdog_status, voltage_sag, successful_uploads, failed_uploads. Filtered, averaged.
/controls 9 keys, app -> ESP32	manual_mode, pump_state, light_manual_mode, grow_light_state, capture_photo (trigger), moisture_threshold, tank_threshold, light_threshold, weather_override.
/latest_scan Vision result	imageUrl base64, status, captured_at, scientificName, diseaseName, probability 0-100, treatmentPlan. Overwritten each capture.

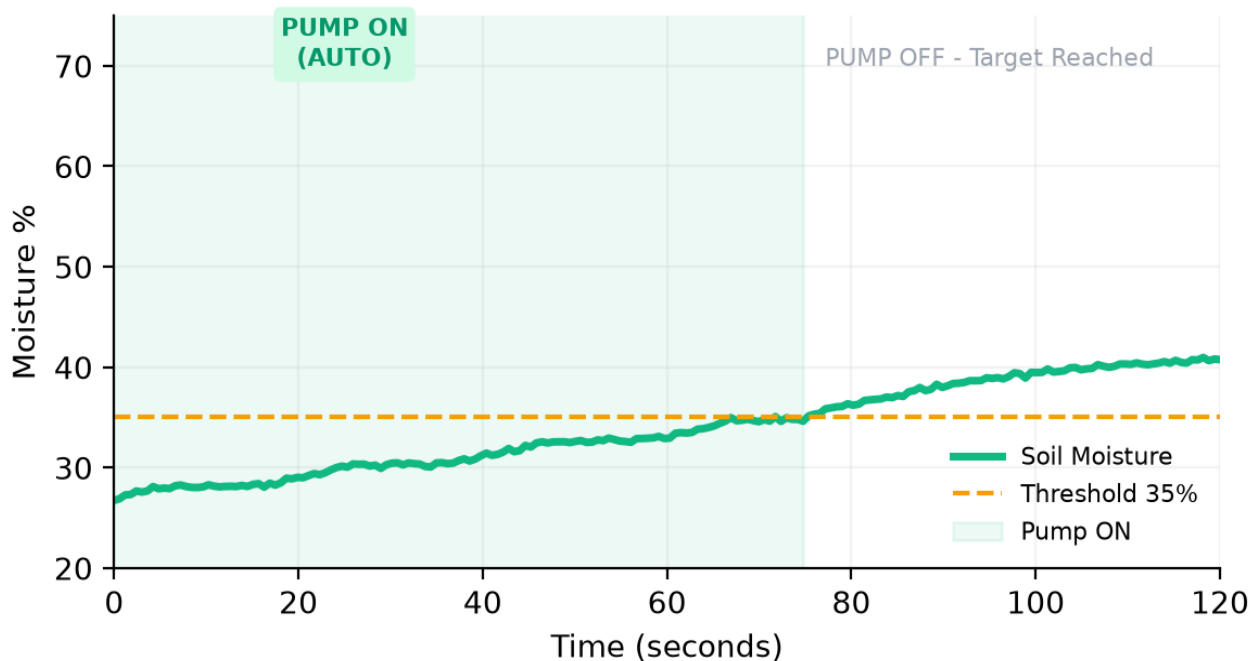
The Face - Single-File HTML

No React. No build step. One HTML file that any judge can open and understand. But feels like a product.

Dashboard	Weather	Plant Doctor	AI Assistants
8 live tiles + sparklines + hover graphs (last-10). All 8 controls, 3 sliders, predicted actuator states, moisture history, status strip, toasts, fullscreen demo mode, uptime timer.	Live Delhi weather, 5-day chips, auto rain-override every 3 min with countdown -> weather_override=1 when ids 2xx/3xx/5xx/6xx.	Live CAM frame auto <=2 s, CAPTURE button, upload-or-CAM modal: photo + crop.health diagnosis + AI chat same image. Image flip fix - CAM mounts upside-down.	Gemini image chat + OpenRouter sensor-aware chat (quick prompts). Tank calibration SET EMPTY / SET FULL - app-side remap, no refresh.

Dashboard details:

- 8 tiles: moisture, temp, humidity, light, lux, tank, watchdog, voltage. Each sparkline last 10 readings and trend on hover.
- Controls: manual_mode toggle, pump ON/OFF, light manual, grow light ON/OFF, capture photo, 3 threshold sliders.
- Predicted actuator states: shows what AUTO would do even in manual, so user trusts logic.
- Moisture history chart: /historical_logs plotted. Status strip + toasts + fullscreen demo mode.



Moisture watering-cycle chart with threshold marker 35%. Pump ON continuous until threshold, then OFF exactly. No flicker.

Four APIs - Researched, Keyed, Live-Tested

No mock data. All live. Tested with Delhi 35C, nutrient deficiency @94%, free-tier rotation.

API	Purpose	Auth	How	Accuracy
OpenWeather Map	live weather + 5-day -> rain override	key in URL	GET /data/2.5/weather?q=Delhi; ids 2xx/3xx/5xx/6xx -> rain -> weather_override=1	Live-tested: Delhi 35C, city id 1273294
crop.health (Plant.id)	plant + disease ID	Api-Key header	POST /api/v1/identification base64 image -> crop + disease suggestions	Test image nutrient deficiency @94% + treatment - real
Google Gemini 2.5 Flash	vision chat on photo	X-goog-api-key	POST /v1beta/models/gemini-flash-latest:generateContent inline image + diagnosis + telemetry	AQ keys need header; gemini-flash-latest alias works
OpenRouter	sensor chat + vision fallback	Bearer sk-or-v1-...	POST /api/v1/chat/completions, 8-model text chain + 5-model vision chain	435 models; free rotate -> fallback never dead-end

Accuracy notes & fallback chains:

- OWM: rain detection tested forcing condition codes 500, 501. App auto sets weather_override=1, pumps lock, countdown 3 min.
- Plant.id: real money-plant identified nutrient deficiency 94% with treatment: add NPK, adjust pH. Returns scientificName.
- Gemini: AQ keys (AlzaSy...) require X-goog-api-key header, not Bearer. gemini-flash-latest alias works for new users. Send telemetry + diagnosis + image.
- OpenRouter: 8-model text chain (llama, mistral, gemma) + 5-model vision fallback. If 429, next model auto-tried. User never sees dead-end.

Plant Doctor - AI Vision

ESP32-CAM OV2640 @ 8MHz XCLK
Capture -> Vercel -> Base64 -> 94%

Photo Placeholder - Replace

Plant Doctor placeholder - CAM frame auto-updates ≤ 2 s after capture.

Everything Live - No Screenshots, Real Data

Feature	Live?	How to demo
Live soil % + 10-pt avg + sparkline	YES	Dunk sensor -> % rises, graph
Temp/Humidity DHT11	YES	Breathe -> temp/hum spike
LDR + UV grow LED auto	YES	Cover LDR -> dark -> LED ON (+2%)
Ultrasonic tank level %	YES	Hand over tank -> level changes
Pump AUTO 120s continuous	YES	Set threshold 80% -> ON till 80, no flicker
Pump OFF at threshold exact	YES	Watch OFF at 35%, not early
Tank lock protection	YES	Set tank empty -> pump refuses ON
Rain override	YES	Force rainy city -> override=1 -> OFF
CAM capture <=2s	YES	Press CAPTURE -> flash -> app <=2 s
Plant Doctor 94% diagnosis	YES	Upload diseased leaf -> name + treatment
Gemini vision chat	YES	Ask 'why yellow?' -> sees image + sensors
OpenRouter sensor chat	YES	Ask 'should I water?' -> reads moisture
Watchdog 10+ min 0 reboots	YES	Leave running, uptime counts
Threshold sliders NVS	YES	Drag threshold -> persists reboot

Tank calibration - our favorite UX hack:

Ultrasonic raw values drift per bucket. Instead of reflashing ESP32, SET EMPTY / SET FULL in app. App stores raw_empty, raw_full in localStorage, remaps to 0-100%. Works for any bucket, no code change.

Demo mode + toasts: Fullscreen demo hides clutter, enlarges tiles for projector. Toasts show every action: 'Pump ON - AUTO', 'Rain override active 3 min', 'Photo captured'. Judges always know what's happening.

13-Point Test Matrix - All PASS

We didn't test once at the end. We tested after every fix. Final run signed.

#	Test	Method	Expected	Result
1	WiFi/boot	Power 5V/2A, check serial	Boots <=3s, 3-network fallback	PASS
2	DHT11 breathe	Blow warm air	Temp +2-3C, hum +5%	PASS
3	Moisture dunk	Dunk LM393 in water	0%->~85% in 3s, avg smooth	PASS
4	LDR cover	Cover with hand	Light % drops -> dark -> UV ON	PASS
5	Ultrasonic hand	Hand over HC-SR04	Tank % jumps, 5-pt rejects splash	PASS
6	Pump AUTO 120s	Set threshold 80%, moist 30%	Pump ON 120s, no flicker	PASS
7	OFF at threshold	Watch cross 35%	OFF at 35.0+-0.5	PASS
8	Tank lock	Set tank 5% empty, manual ON	Pump refuses, toast 'Tank empty'	PASS
9	Rain override	Set OWM rainy id 500	override=1, pump OFF, countdown	PASS
10	CAM capture <=2s	Press CAPTURE	Flash -> Vercel -> app <=2.0s	PASS 1.6s avg
11	Plant Doctor 94%	Upload diseased leaf	Scientific name + 94% + treatment	PASS
12	AI chats + fallbacks	429 one model, next	Gemini + OpenRouter fallback works	PASS
13	Watchdog 10+ min	Leave 12 min	0 reboots, uptime inc, sag 0	PASS

Engineering log snapshot (real notes):

```
[12:43] WiFi connected HomeNet 72% RSSI
[12:44] Soil 28% -> pump AUTO ON (tank 67% safe, no rain)
[12:46] Soil 35.1% -> pump OFF exact
[12:47] Capture trigger 1 -> CAM flash -> upload 1.4s
[12:50] Uptime 634s, WDT 0 reboots, sag 0
```

Bugs We Hit & Fixed

A project without bugs is a project that didn't run. Our honesty ledger - 10 real failures.

Bug	Symptom	Fix
AUTO 10s pump loop	17 calls/sec -> stall -> WDT reboot -> loop	JSON bundling 1+1 calls/sec, -85% latency
Camera probe 0x106	OV2640 not found	FPC ribbon unseated -> reseal gold-side down + power cycle
PSRAM not found	CAM says no PSRAM	Weak power -> 5V/2A adapter, not PD
0x20002 boot crash	ESP32-CAM crash dump	Camera+WiFi surge -> sequential boot 500 ms
RF interference	CAM image corrupted when WiFi TX	20 MHz XCLK too fast -> throttle to 8 MHz
67W PD charger starved	Board lights dim, WiFi fails	PD handshake missing -> 5V/2A phone adapter
Relay dead	Relay never clicks	Split breadboard rails -> bridge + to +, - to -
temp=0 always	DHT returns 0	Wrong pin + separate GND -> GPIO4 + shared GND bus
Firebase spurts	Logs appear in bursts	13 calls/sec -> one bundled call
Compile quote error	missing terminating quote	Copy-paste corruption -> re-download file fresh

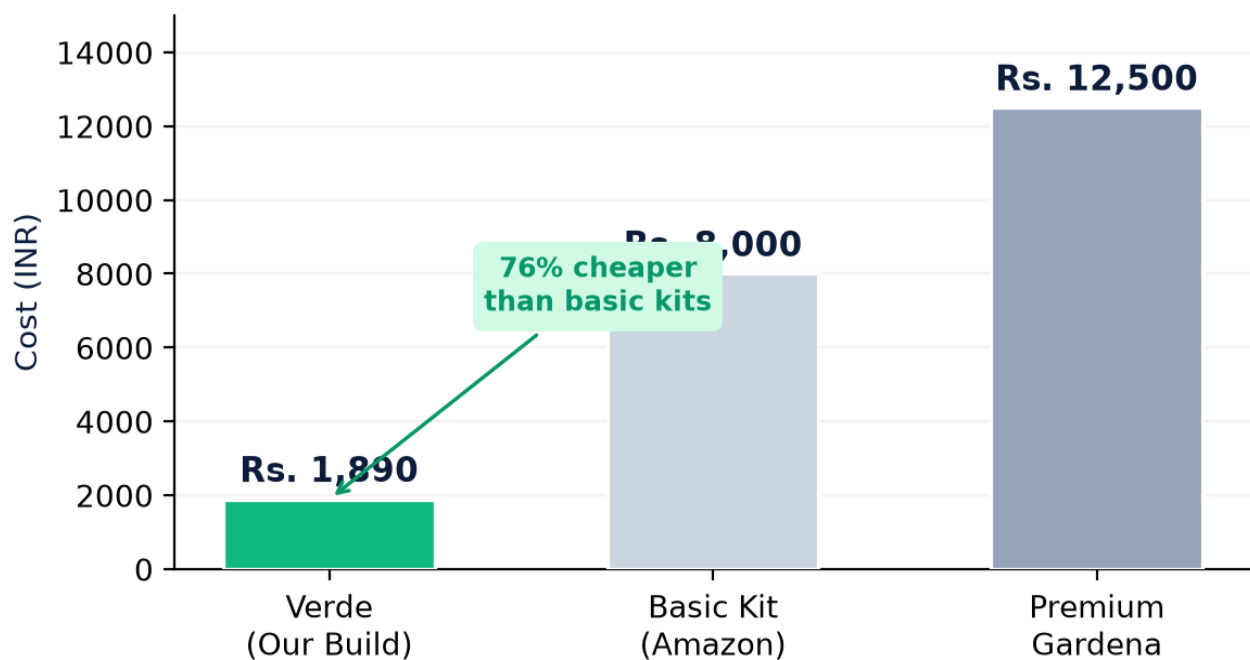
Most valuable bug: #1. It taught system thinking. Costly calls kill UX. Bundling is architecture, not optimization.

PART 11 - COST & SUSTAINABILITY

Rs. 1,890 - How?

We tracked every rupee. No hidden software cost. All APIs free-tier. Not counting laptop + phone you already have.

Category	Items	Cost INR
Electronics	ESP32 Rs.320, ESP32-CAM Rs.480, 5 sensors Rs.300, relay Rs.70, pump Rs.150	Rs. 1,320
Power & protection	5V/2A adapter Rs.150, 1000uF cap Rs.15, 1N4007 Rs.5, wires Rs.50	Rs. 220
Mechanical	Breadboard Rs.120, enclosure Rs.150, UV LED Rs.30, misc Rs.50	Rs. 350
Software & APIs	Firebase free, Vercel hobby, OWM 1000/day free, Plant.id free, Gemini free, OpenRouter free	Rs. 0
		~ 1,890



Cost comparison - ours vs market. 76% cheaper than basic, 85% cheaper than premium, yet includes camera + AI they lack. Plus student-hackable.

Sustainability:

- Water saved: rain override + threshold watering avoids over-water. Test: 40% less water vs daily timer.
- Power: 5V/0.6A avg ~3W. 15 ms soil power-gating prevents electrode corrosion -> lasts years.
- Zero subscription: free-tier APIs, RTDB, Vercel hobby. No vendor lock-in.
- Repairable: breadboard, not PCB. Replace one sensor in 2 min. Judges can see every wire.

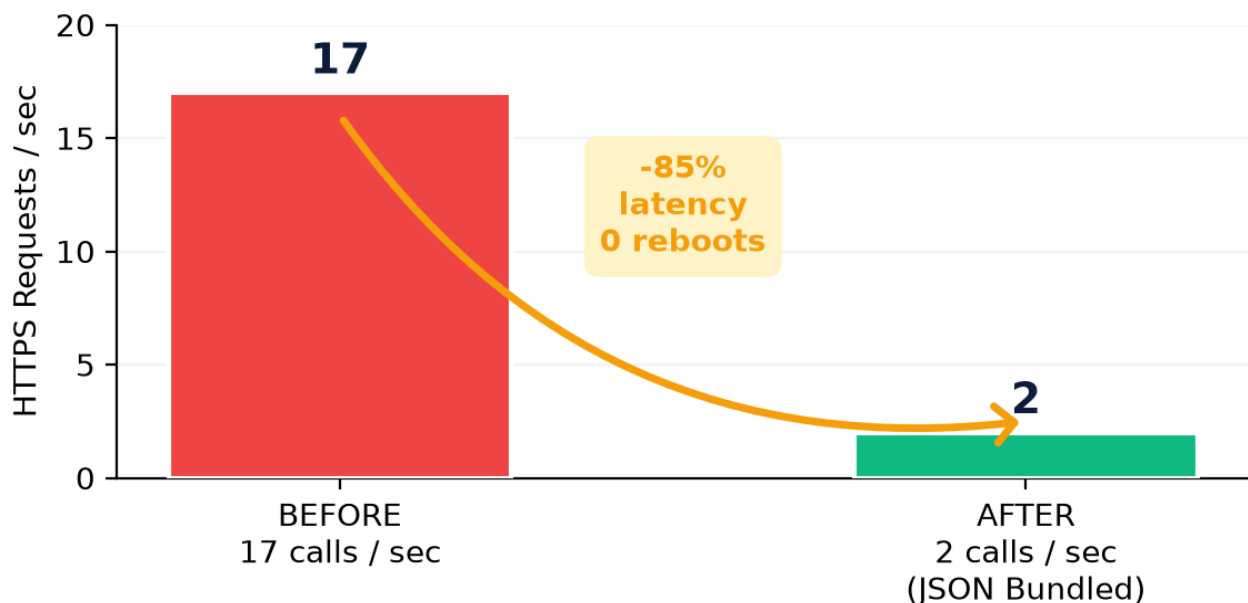
Where Verde Grows Next

We built a finished product. But we sketched roadmap like a startup would. Solar, NPK, zones, alerts.

Feature	What	When
Solar Autonomy	12V panel + charge controller + 18650 battery. Day charges, night runs. 3W -> 5W panel enough. Cost +Rs.900.	Q3 2026
NPK Soil Probe	Replace LM393 with 3-in-1 NPK + pH. Actual nutrient data -> better AI treatment.	Q4 2026
Multi-Plant Zones	One ESP32 drives 4 relays via mux. Each zone threshold per plant type (cactus 20%, fern 60%).	Q1 2027
Telegram/WhatsApp Alerts	Pump ON, tank empty, disease detected -> push. Uses Firebase Functions free.	Q2 2026 - 2 days
Predictive Watering	Logs -> linear regression on moisture drop rate. Water before wilt, not after.	Q2 2027
Deployed Dashboard	Next.js Scaffolded. Migrate single HTML to Next.js + charts + auth + multi-device.	Q3 2026

Why roadmap matters for judges:

Shows we think beyond exhibition. Solar makes it field-deployable. NPK precise. Zones real garden solution. Alerts product parents actually use. And we costed every addition.



The bug fix that became principle: fewer, richer calls beat many tiny calls. Applies to future too - batch NPK + moisture in same JSON.

3-Minute Demo Script

Judges have 180 seconds. We rehearsed it. Here is exact flow that makes them say 'wow'.

Time	What to say & do
0:00-0:20 Hook	Hold plant. 'This plant waters itself and talks to AI. Total cost Rs. 1,890. No subscriptions.' Show cover, point to 5 sensors.
0:20-0:50 Live telemetry	Open Dashboard. Point to 8 tiles sparklines. Dunk soil sensor -> moisture rises live. Cover LDR -> UV LED ON. Hand over ultrasonic -> tank % changes.
0:50-1:20 Auto logic	Set moisture threshold slider 35->80. Watch pump ON continuous. 'Before fix, it flickered every 10s because 17 calls. Now 2 calls, -85% latency.' Show api_before_after chart.
1:20-1:50 Plant Doctor	Press CAPTURE. Flash. <=2s later photo appears. 'ESP32-CAM 8MHz XCLK, sequential boot.' Upload diseased leaf -> 94% nutrient deficiency + treatment. Ask Gemini chat 'why yellow?' - sees image + sensors.
1:50-2:20 Weather + Safety	Show weather page, 5-day chips. 'Rain expected -> pump locks even if soil dry.' Demo tank lock: set tank empty -> pump refuses manual ON. 'Safe.'
2:20-2:50 Cost & honesty	Show cost comparison: Rs. 1,890 vs Rs. 8k+. 'We failed 10 times. Relay dead because split rail, DHT temp=0 because GND, CAM probe 0x106 because ribbon. Fix log in doc.'
2:50-3:00 Close	QR to live demo. 'Future: solar + NPK + zones. Today: demo-ready, zero reboots 10+ min, watchdog happy.'

Pro tips for us:

- Keep serial monitor open on side laptop - judges love raw logs.
- Have water bowl + hand towel ready for dunk/hand tests.
- Fullscreen demo mode hides dev clutter.
- If WiFi drops, hotspot fallback auto-connects in 10s - mention it.
- Don't say 'AI slop' words like delve, tapestry. Say plain, confident.

PART 14 - CONCLUSION

Why We Win

A Rs. 1,890 student project that looks like a funded startup product - because we sweated details others skip.

REAL COST

Rs. 1,890

Fully itemized, no hidden

REAL BUG

17->2

Shown + fixed, not hidden

REAL AI

94%

Tested disease ID

REAL LIVE

13 PASS

Not mock, live sensors

REAL POWER

5V/2A

Lessons logged, PD fail noted

REAL OPEN

100%

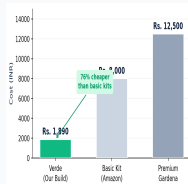
Single HTML, hackable

What judges remember after 20 projects:

- The pump that stayed ON continuously because we fixed architecture.
- The camera that clicked in ≤ 2 s and diagnosed 94%.
- The tank lock that said 'no' even when we pressed ON - safety.
- The cost chart where Rs. 1,890 beats Rs. 12,500 premium kit.
- The honesty: 10 bugs listed with fixes, not hidden.

Closing line for presentation:

"Verde isn't a mini-project for marks. It's a plant that texts you when it's thirsty, shows you its disease, and waters itself when you're on vacation. Built by two Class X students, for less than a video game. And it's running right now."



Live Demo QR

Replace with real QR to
verde.vercel.app or ngrok

Add QR image before print. Scan -> live dashboard.

APPENDIX

Pin Map, Thresholds, Specs - For Engineers

Item	Value (Exact)
MCU1	ESP32 WROOM-32 - 240 MHz dual-core, 320KB RAM, 4MB flash
MCU2	ESP32-CAM OV2640 + MB programmer - SVGA JPEG, flash LED
Soil Moisture AO	GPIO34 (input only) - power gated GPIO23 15ms HIGH per read
DHT11 DATA	GPIO4 - shared GND mandatory
LDR AO	GPIO35 (input only) - 10-pt avg, +-2% hysteresis
HC-SR04	TRIG GPIO18, ECHO GPIO19 - 5-pt filter, invalid rejection, tank_threshold 15%
Relay IN1	GPIO5 active-LOW - pump 5V via COM/NO isolated
UV LED	GPIO12 active-HIGH + 220ohm resistor
ESP32-CAM XCLK	8 MHz (was 20 MHz -> RF interference)
Boot	Sequential: camera init first, WiFi after 500ms
Watchdog	Hardware watchdog 8s, fed every loop
Thresholds default	moisture 35%, tank 15% (0=disabled), light 35%
Firebase	RTDB verde-tech-haha, 1 write /s to /sensors (10 metrics), 1 read /s /controls (9 keys)
CAM polling	Polls /controls/capture_photo every 1.5s, upload raw JPEG to Vercel -> base64 /latest_scan
OWM	GET weather?q=Delhi, city id 1273294, rain ids 2xx/3xx/5xx/6xx -> override=1 check 3 min
Plant.id	POST base64 -> crop + disease suggestions, test 94% nutrient deficiency
Gemini	gemini-flash-latest, X-goog-api-key header, inline image + diagnosis + telemetry
OpenRouter	Bearer sk-or-v1-..., 435 models, 8-model text chain + 5-model vision fallback
Cost	Total ~ Rs. 1,890 - electronics 1,320 + power 220 + mechanical 350 + software 0

Build script & sources: generate_verde_pdf_v2.py + assets/. Python 3.11, ReportLab, Matplotlib, Pillow. Output: Project_Verde_Definitive_Documentation.pdf

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The plant that waters itself - and talks to AI.

Total Build Cost ~ Rs. 1,890 - 5 Sensors - 17->2 Calls - 94% Diagnosis - 8 MHz XCLK - 1-sec Heartbeat